

a What was measured

Muller et al. 2016

19 intracellular amino acids (cysteine omitted) by LC-SRM/MS, in mM.
Prototrophic BY4741 deletion collection, synthetic minimal medium, 30 °C.
Reference: robust mean over all strains.

AminoAcidMuller2016Dataset

Cooper et al. 2010

17 amine peaks by CE-LIF, some holding two or three amino acids, as a ratio to the plate mean.
MATa deletion collection, synthetic complete medium; temperature not stated. Reference: 1.0.

AminoAcidCooper2010Dataset

Cachera et al. 2023

CRI-SPA moves the Btx cassette (CYP76AD1, DOD, ARO4 K229L, ARO7 G141S) into each deletion strain; colony yellowness score, centered at 0.
Screen plate YPD + G418 (the record stores SC, 30 °C; see Caveats).

BetaxanthinCachera2023Dataset

b The record

Experiment

genotype.perturbations: one KanMX deletion; Cachera adds the four cassette genes as gene_addition

environment: media (name, state), temperature

phenotype (MetabolitePhenotype): metabolite_level {key: value}, n_replicates, metabolite_level_se, measurement_type

Reference

genome_reference: strain BY4741

phenotype_reference: robust mean, 1.0 or 0 per key

Publication: DOI and PubMed id

All three are MetabolitePhenotype records with graph_level 'metabolism'; keys and units differ by dataset.

c Query and aggregation

amino_acid_betaxanthin.cql: the three datasets; every deletion in \$gene_set, at least one deletion; gene additions pass

Neo4jQueryRaw: one entry per Experiment node

GenotypeAggregator (no converter, no deduplicator): one record per perturbed gene set

Muller

Cooper

deletion of gene X alone

Cachera

gene X plus the four cassette genes: a different gene set, so a record of its own